NEURAL NETWORKS Explained

No jargons Just clarity

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WHAT IS NEURAL NETWORKS?

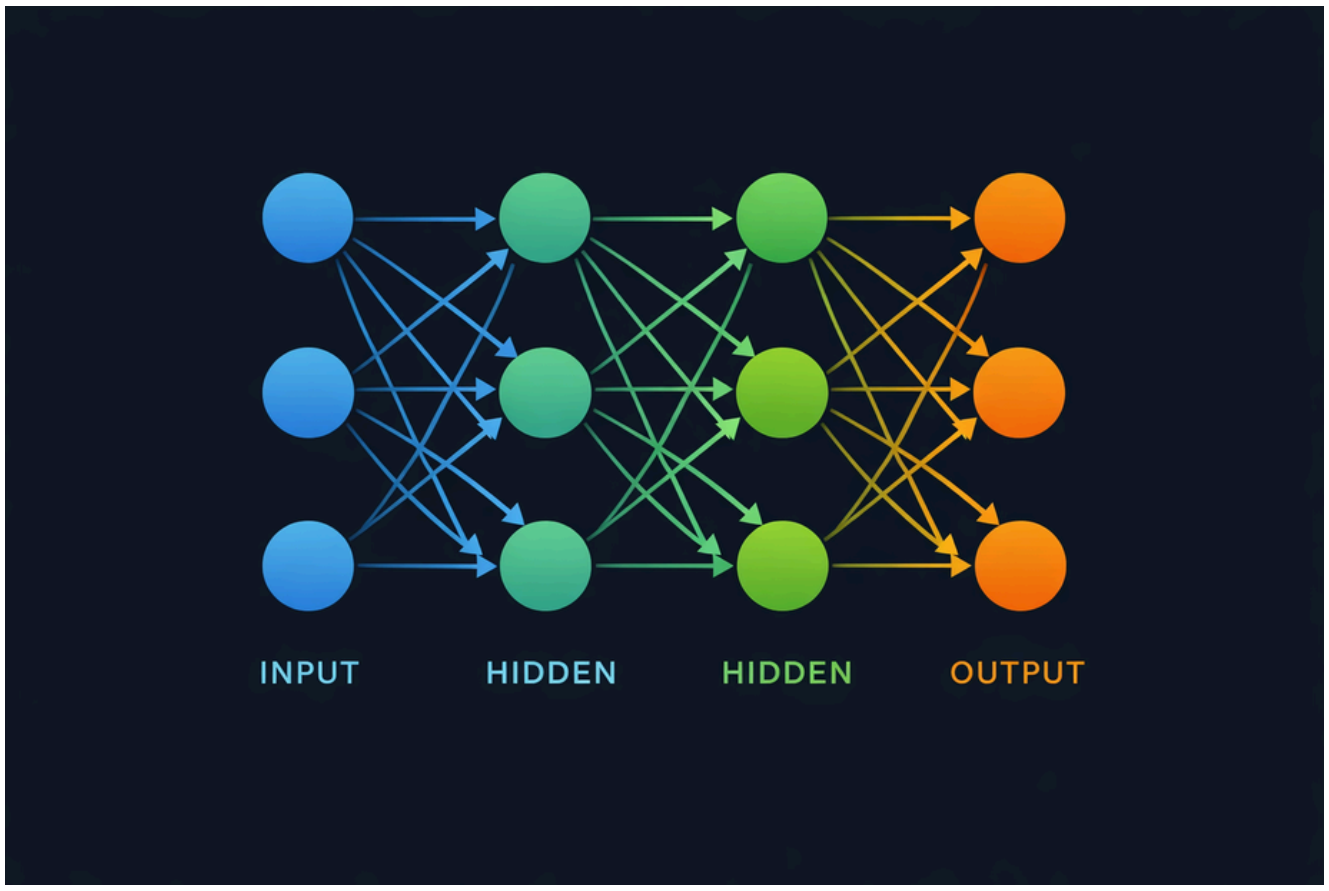
- Neural networks are a type of **machine learning model** inspired by how the human brain works
- Learns patterns from data to make predictions

STRUCTURE OF NEURAL NETWORK

Neural network is made up of layers of connected nodes (called neurons):

- **Input layer** – receives data (like images, text, or numbers)
- **Hidden layers** – process the data through calculations
- **Output layer** – produces the final result (like a prediction or classification)

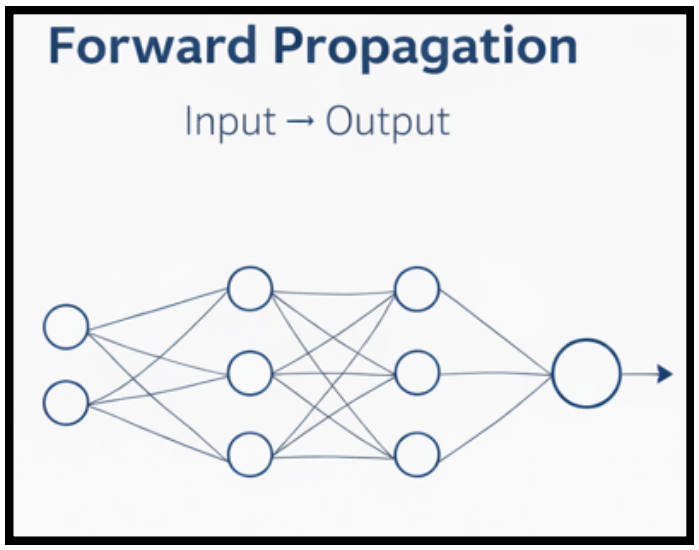
Each connection between neurons has a weight, which the network adjusts during training to improve its accuracy.



HOW IT LEARNS

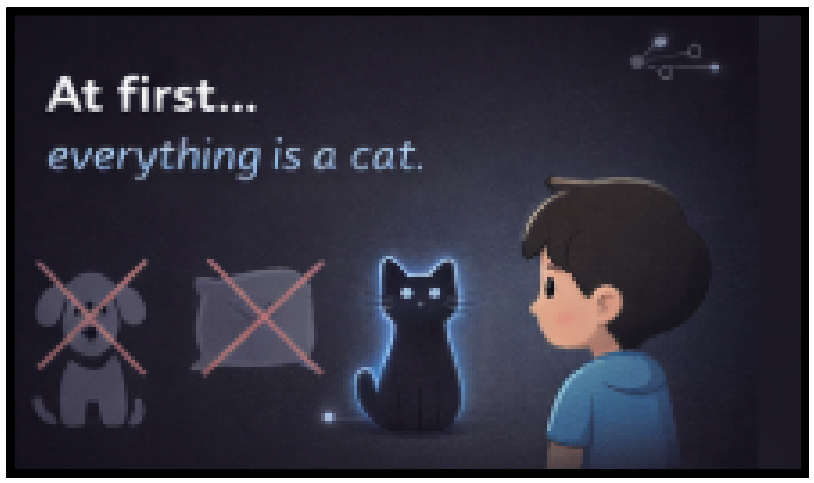
FORWARD PROPOGATION

- Data flows from input → hidden → output
- Each neuron applies weights, bias, and activation function
- Produces an output prediction



Example:

Imagine teaching a child to recognize a cat



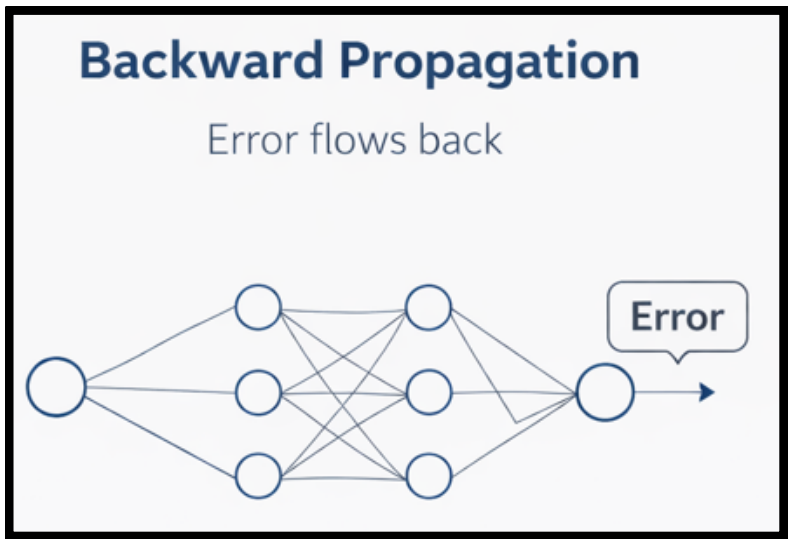
HOW IT LEARNS

ERROR CALCULATION

- Compare prediction with actual result (error)

BACKWARD PROPOGATION

- Error is propagated backward through the network
- Weights and biases are adjusted using optimization (e.g., gradient descent)
- Goal: Minimize error and improve accuracy

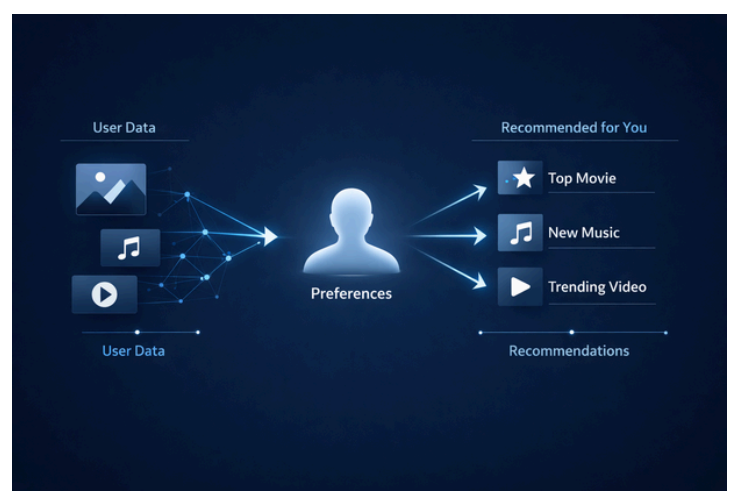


LEARNING PROCESS THROUGH PATTERNS

APPLICATIONS

Neural networks power many modern technologies, such as:

- Image and speech recognition
- Chatbots and language models (like chatgpt)
- Self-driving cars
- Recommendation systems (like Netflix or YouTube)



If this made Neural Networks simple for you, save it for later.
Stay tuned for more simple tech breakdowns